

Simra Ali

Computer Engineering · Robotics, Autonomous Systems & Embedded Design

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#1 in MA · #11 in the world FTC 3565, out of 7,800+ teams (2026)	3× 1st place Hackathons & CAD competitions, 2024–25	GPA 4.18 · SAT 1560 790 Math · 770 EBRW	MIT BWSI · Harvard SSP Autonomous systems coursework
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EDUCATION

Westford Academy · Westford, MA GPA 4.18 · SAT 1560 (790 Math, 770 EBRW) · Academic High Honors, 2023–present Coursework: AP Calculus BC, AP Physics C, AP Chemistry, AP Computer Science A, AP Biology, AP U.S. History, AP Psychology, AP Microeconomics	Class of 2027
MIT Beaver Works Summer Institute · Lincoln Laboratory Autonomous Air Vehicle Racing (Onsite) · Basics of ASICs · Microelectronics & Hardware Development · Python Core · Version Control: Git/GitHub & UNIX	Feb 2024 – Aug 2026
Harvard Summer School · Cambridge, MA Robotics, Autonomous Vehicles, Drones, and Artificial Intelligence — Grade: A	Summer 2025

ROBOTICS LEADERSHIP

FIRST Robotics Competition · Stormgears 5422 (45 members) Co-Captain · Lead CAD / Mechanical Designer · Drive Team Coach · Former Electrical Lead Taught myself SolidWorks after the team's experienced designer graduated, then helped lead a custom-robot effort that earned the FIRST Quality Award for design intent, robustness, and fabrication quality. - Designed competition mechanisms in SolidWorks — chassis, intake, hopper, shooter, electrical board, belly pan, and component holders — specified for serviceability between matches. - As Electrical Lead, ran summer and fall training, managed inventory, scheduling and task tracking (Slack, Trello), and trained the student who succeeded me in the role. - Drive Team: Coach at Rebuilt 2026 and Technician at 2025 New England District Championship; Driver at the 2024 Clipper Clash. - Presented to sponsors including IBM, Triton Systems and SolidWorks.	2023 – Present
FIRST Tech Challenge · Ghost Robotics 3565 (10 members) Vision Software · Onshape CAD 2026 Massachusetts State Champion and Winning Alliance Captain; Ross Division Winner at the FIRST World Championship — team ranked #1 in Massachusetts and #11 in the world out of 7,800+ teams. - Work deliberately across vision software, CAD, strategy, media and outreach rather than limiting myself to one formal title.	2023 – Present

ENGINEERING PROGRAMS & RESEARCH

Autonomous Air Vehicle Racing · MIT Beaver Works Summer Institute, AeroAstro Four-week onsite program - Designed, programmed, repaired, tested and raced autonomous drones; studied computer vision, state estimation, control systems, navigation and flight dynamics. - Explored socially aware path planning for autonomous drones and connected with an MIT PhD researcher working in the field. Honorable Mention, BWSI “You Belong in STEM” Essay Contest.	Summer 2026
Robotics, Autonomous Vehicles, Drones & AI · Harvard Summer School Grade: A - Studied sensors, motion planning, SLAM, reinforcement learning, autonomous systems and computer vision. - Co-led the “Solve It, Defend It!” innovation club and developed Flood Watch, a concept for predicting flood risk from drone and autonomous-vehicle data.	Summer 2025

Women in Technology Program · BAE Systems

Jan – May 2026

- Hands-on labs, workshops, tours and presentations spanning electrical, microwave/RF, software, mechanical, signal processing, optical, manufacturing and failure-analysis engineering.

Summer Immersion & Pathways Programs · Girls Who Code

Jun 2024 – Jul 2025

Cybersecurity, cryptography, data science, machine learning, game design and web development.

PROJECTS & COMPETITIONS

The Bat · GirlHacks at Microsoft NERD — 1st Place

Jun 2025

- Assistive navigation device for blind and low-vision users: ultrasonic sensors detect obstacles and a buzzer returns auditory feedback. Designed, built and demonstrated within the hackathon.

HireAble · NetScout Tech for Good — 1st Place

Mar 2025

- Web platform connecting people with disabilities to inclusive employers, integrating database and web technologies to make hiring more accessible.

VTXCADs CADathon · Phillips Exeter Academy Robotics — 1st Place

Dec 2024

Firefly · Girls Who Code game design project

Jul – Aug 2025

- Interactive game built with the p5play / p5.js JavaScript libraries, using **TensorFlow.js hand-gesture detection** as the input method.

SolidWorks Demonstration Robot · built for team sponsor

May – Sep 2024

- FRC-style robot with autonomous vision to locate tennis balls, an intake to collect them, and a shooter to launch them at a target.

OUTREACH, MENTORSHIP & LEADERSHIP

Outreach Coordinator · Middlesex County Robotics Club, Boxboro, MA

Aug 2023 – Present

- Organize and lead Girl Scout STEM Fest, FIRST LEGO League trainings and scrimmages, the Apple Blossom Parade, and elementary and middle school STEM events.
- Mentored an all-girls FTC team** and led the upgrade of the SaturnReach outreach robot; emphasis on getting girls into technical roles and keeping them there.

Co-Founder & Co-Lead · Women in Computer Science, Westford Academy

Sep 2024 – Present

- Teach Python, support member projects, run hackathon preparation, and share STEM opportunities that are easy to miss if nobody tells you they exist.

Manager · Westford Academy Broadcasting Club

Sep 2024 – Present

- Coordinate club operations while working the shows as anchor and production crew: teleprompter, sound board, video switching, scripts and school-wide broadcasts.

Cashier · Demoulas (Market Basket), Westford, MA

Jul 2023 – Dec 2024

HONORS & AWARDS

FIRST Leadership Award (Dean's List) Semi-Finalist	2026	Academic High Honors, Westford Academy	2023–26
BWSI "You Belong in STEM" Essay — Honorable Mention	2026	AP Scholar Award	2026
GirlHacks at Microsoft NERD — 1st Place	2025	FRC Quality, Team Spirit & Gracious Professionalism	2025–26
NetScout Tech for Good — 1st Place	2025	FTC Control, Inspire, Innovate & Connect	2025–26
VTXCADs CADathon — 1st Place	2024	Diversity, Equity & Inclusion Award, Westford	2023
Toastmasters Youth Leadership Award & Medal	2021	Honored Citizen Award, Westford Public Schools	2023

TECHNICAL SKILLS

CAD / Mechanical SolidWorks, Onshape, 3D printing, Autodesk Tinkercad

Hardware Arduino, wiring, soldering, crimping, electrical-board layout, sensors, systematic troubleshooting

Software Python, Java, JavaScript, HTML/CSS, SQL, Git/GitHub, WPILib, TensorFlow, NumPy, p5.js

Robotics / AI Computer vision, SLAM, motion planning, reinforcement learning, autonomous systems, state estimation